Human Resource Management System

Course: Software Engineering | Semester: Spring 2026 | Submitted by: Team HRM

1. Project Title

Human Resource Management (HRM) System: A Web-Based Platform for Employee Administration, Payroll Processing, and Attendance Tracking.

2. Introduction

2.1 Overview of the Domain

All kinds of organizations need to have well-planned HR processes to stay productive and follow the rules. Managing people manually through spreadsheets, paper files, and disconnected tools is no longer an effective way to do things. Modern HR platforms put attendance tracking, payroll automation, and employee self-service into one place, which reduces the amount of administrative work and allows for data-driven decisions about the workforce.

2.2 Motivation

This project came from a simple observation: many mid-sized companies have a hard time justifying the cost of enterprise HR software, but their manual workflows create real problems. We want to show how full-stack engineering principles can be used to build an HRM system that is affordable and customizable. We also want to show that this system could genuinely work in a real business context.

3. Background to the Problem

3.1 Current System Overview

Most small and medium organizations track attendance with sign-in sheets or basic biometric devices. These devices don't integrate with any software. Payroll is calculated in Excel, and employees are assigned tasks via email or chat. Employee records are stored across multiple disconnected files or physical folders.

3.2 Limitations

These methods cause ongoing problems. Attendance records are unreliable and easy to fake. Payroll errors happen often and take a lot of time to fix. Managers don't have a central view of who is working on what, and employees can't easily access their own records. There is no record of what has been done, no analytics, and it cannot be made larger.

3.3 Need for a Proposed Solution

A digital platform that's all in one place can fix all of these problems. It can put all of an employee's data in one place, do payroll automatically, let you see if employees are where they're supposed to be right away, and let human resources managers and employees see what they need to see in the system.

4. Problem Statement

Organizations that rely on manual or disconnected HR workflows often face unnecessary problems with tracking attendance, processing payroll, and managing tasks. Without a centralized system, HR teams spend too much time on administrative tasks, errors go undetected until they cause financial or legal issues, and employees can't access their own records. We need a web-based HRM system that is integrated to combine these operations into one platform that's secure and can be tracked.

5. Objectives of the Project

1. Develop a role-based dashboard for HR managers with task creation, to-do lists, and employee management tools.
2. Build an employee onboarding module for adding, editing, and archiving staff profiles.
3. Implement a payroll engine that auto calculates salaries, deductions, and generates printable payslips.
4. Design a clock-in/clock-out attendance system with a GitHub style heatmap visualization of daily activity.
5. Enable employees to view and print their attendance history and payroll summaries from a self-service portal.
6. Implement JWT-based authentication with distinct HR Admin and Employee access levels.

6. Proposed Solution Overview

Built on the MERN stack MongoDB, Express, React, Node.js with RESTful APIs connecting two distinct role-based interfaces:

HR Admin Panel	Employee Portal
✓ Dashboard with task & to-do management	✓ Personal clock-in / clock-out interface
✓ Full employee CRUD (add, edit, archive)	✓ Attendance heatmap (GitHub-style grid)
✓ Payroll configuration & salary processing	✓ Printable pay slips & attendance reports
✓ Attendance reports & analytics dashboard	✓ Self-service profile & payroll history

MongoDB	Express.js	React.js	Node.js	JWT Auth
Database	Backend API	Frontend UI	Runtime	Security

7. Target Users

7.1 Primary Users

HR Managers manage the full employee lifecycle. They also process payroll, monitor attendance, and assign tasks. Employees can clock in and out, view attendance records, and download pay slips.

7.2 Secondary Users

Company directors or finance teams who use analytics dashboards and payroll summaries to make important decisions. System administrators are responsible for creating user accounts and setting up the technology platform.

8. Scope of the Project

8.1 In-Scope

Manage employee profiles, verify employees' identities, and manage HR tasks and to-do lists. Use the clock-in/clock-out system to track employee attendance. Generate payroll calculations and pay slips. Print attendance and payroll reports. Access the admin analytics dashboard.

8.2 Out-of-Scope

Mobile app development, adding biometric hardware, third-party payroll provider APIs (like ADP), AI-driven performance reviews, and support for multiple companies or tenants.

9. Expected Impact and Benefits

Automating attendance and payroll is expected to cut HR processing time by about 65%. Centralizing employee records is expected to completely eliminate the most common type of data error. The self-service portal reduces the amount of back-and-forth between employees and HR for routine requests. The heatmap view gives managers a quick view of how often employees are coming to work. This helps them spot trends in when employees are not coming to work. The platform can be used by 20 to several hundred employees without needing to make any changes to the way it's built.

10. Conclusion

The proposed HRM System solves the problems of manual HR operations being disorganized and inefficient by providing a web-based platform for managing employees, processing payroll, and tracking attendance. It combines practical SE concepts like authentication, database design, API development, and UI engineering in a project that has real world value. The result will be a functional MVP that any growing organization could realistically adopt. It will also be a strong example of the team's full-stack development capabilities.